

Total No. of Questions : 8]

SEAT No. :

PC-2342

[Total No. of Pages : 2

[6354] - 458
B.E. (Civil Engineering)
TRANSPORTATION ENGINEERING
(2019 Pattern) (Semester - VII) (401002)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Use of electronic pocket calculator is allowed.
- 4) Assume suitable data if necessary.
- 5) Neat diagrams must be drawn wherever necessary.

- Q1) a)** The horizontal curve of radius 180 m is having design speed of 60 kmph. The design coefficient of lateral friction is 0.15. **[6]**
- i) Calculate required super elevation if full lateral friction is assumed to develop.
 - ii) Calculate required coefficient of friction if super elevation is not provided.
- b)** Draw a neat cross section of MDR in cutting in rural area. **[6]**
- c)** Define Camber, Shoulder, Kerb, Right of way, Width of formation, and Sight Distance. **[6]**

OR

- Q2) a)** The speeds of overtaking and overtaken vehicles are 80 kmph and 50 kmph respectively on a two way traffic road. The average acceleration during overtaking can be assumed as 0.99 m/s^2 . **[6]**
- i) Calculate safe overtaking sight distance
 - ii) What is minimum length of overtaking zone?
- b)** What is effect of gradient on overtaking sight distance? **[6]**
- c)** What is overturning effects? Explain with a neat sketch. **[6]**

P.T.O.

- Q3)** a) Explain role of bituminous binder in construction of highway. [6]
b) Define Elongation Index (EI). How EI is determined in the laboratory. [6]
c) Explain significance of Marshall method of bituminous mix design. [5]

OR

- Q4)** a) Explain in brief types of tests to be carried out to select the suitable grade of bitumen. [6]
b) What is Angularity number? Give its significance in highway construction. [6]
c) What is significance of aggregate gradation in design of non bituminous layer of flexible pavement? [5]

- Q5)** a) Explain basic concepts in analysis of various stresses in Rigid Pavements. [6]
b) Differentiate temperature stresses and wheel load stresses. [6]
c) What is dowel bar in rigid pavement? Explain its role with respect to functioning stresses. [6]

OR

- Q6)** a) Classify different types of joints in rigid pavements and mention objectives of each. [6]
b) What are the factors causing warping stresses in cement concrete pavements? Explain. [6]
c) Differentiate rigid pavement and flexible pavement with a neat sketch. [6]

OR

- Q7)** a) Discuss various factors that engineer will consider in site selection for a bridge on a major river? [8]
b) Define Abutment. State the various types of abutments. [4]
c) What are the requirements of an ideal permanent way? [5]

OR

- Q8)** a) What are the advantages and disadvantages of temporary bridges? [8]
b) Differentiate between skew bridge and submersible bridge. [4]
c) What are the types of R.C.C bridges? Draw sketch of any one with neat labeling. [5]

